



A Maintenance Program for a bus fleet

Lessons from a Potential CDM Project
in India

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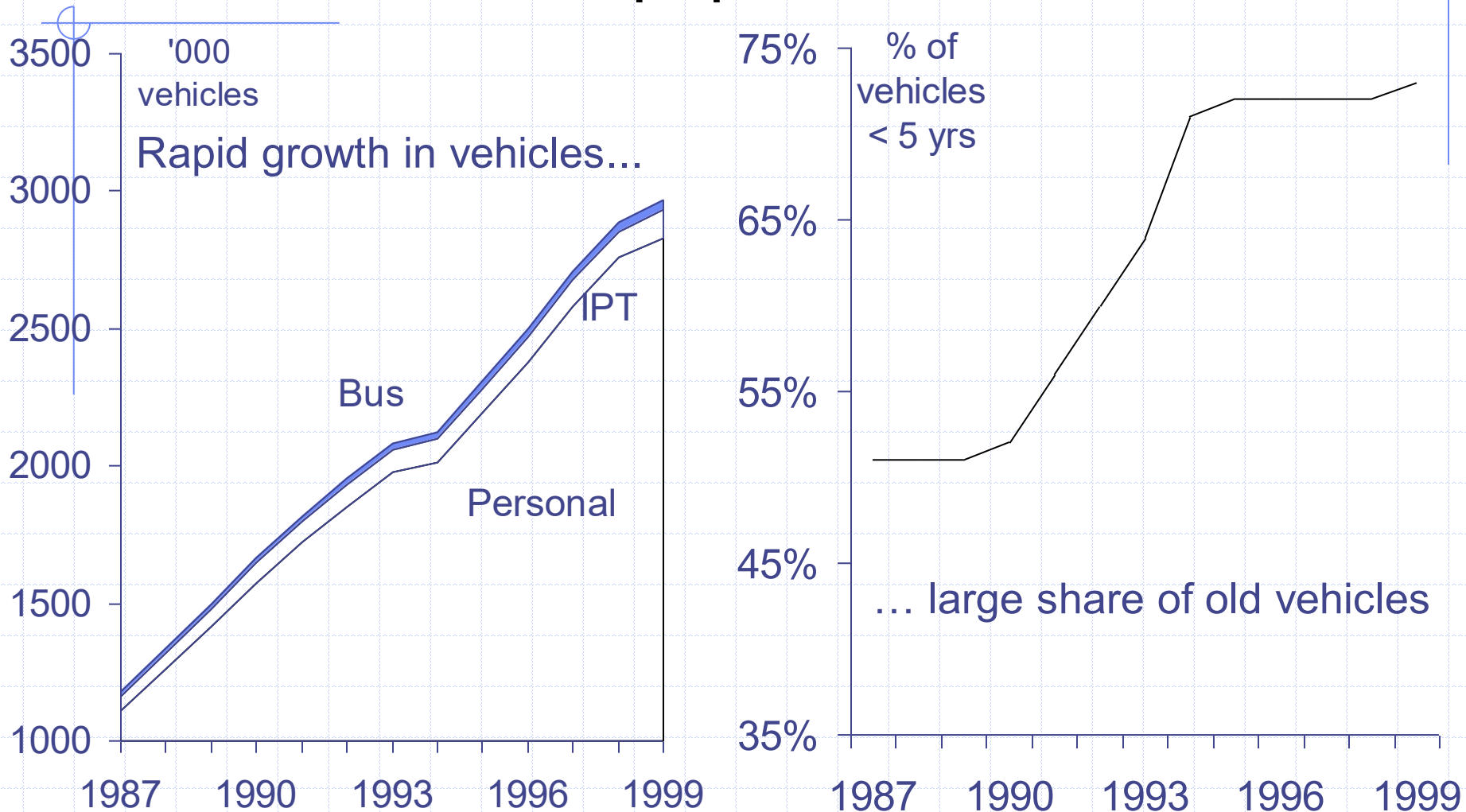
Structure of the presentation

- ◆ The policy context
- ◆ CDM and transport: Synergies
- ◆ Key barriers
- ◆ The Project
 - *Need for an effective maintenance*
 - *Inspection Systems: CDM project*
 - *Emission savings*
 - *Cost estimates*
- ◆ Lessons

Urban transport in India

- ◆ Large urban population
 - *Over 1/3 in 2001*
- ◆ High vehicle density in urban areas
 - *33% concentrated in cities with 10% population*
 - *82% are personal vehicles*
- ◆ Inadequate public transport
 - *Effective public transport in only 10 cities*
 - *Rail in only 3 cities*
- ◆ No urban transport policy
- ◆ Diffused institutional arrangements
- ◆ Implications for economic sustainability
 - *Cities are vertices of continued growth*

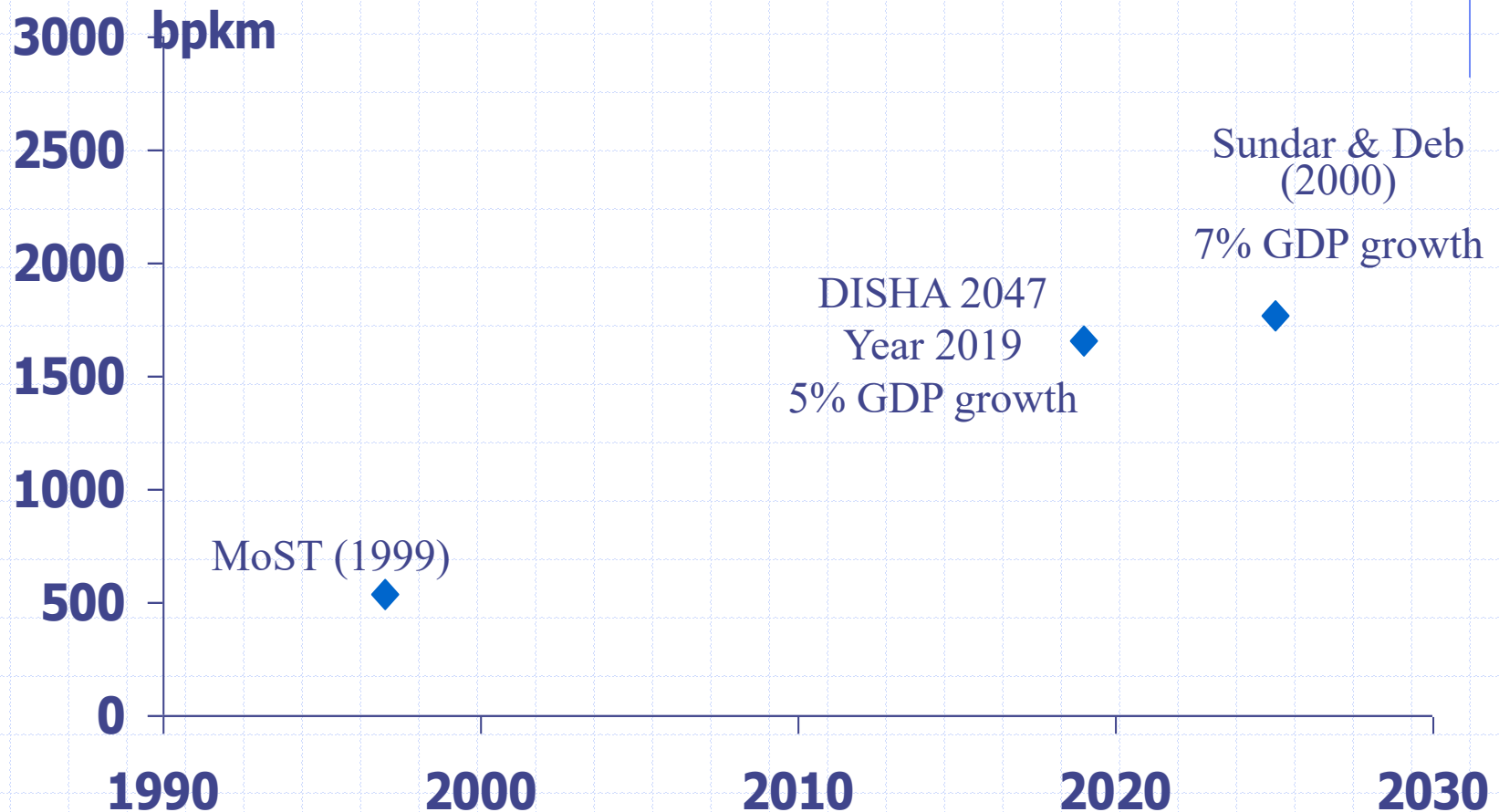
Trends in vehicle population



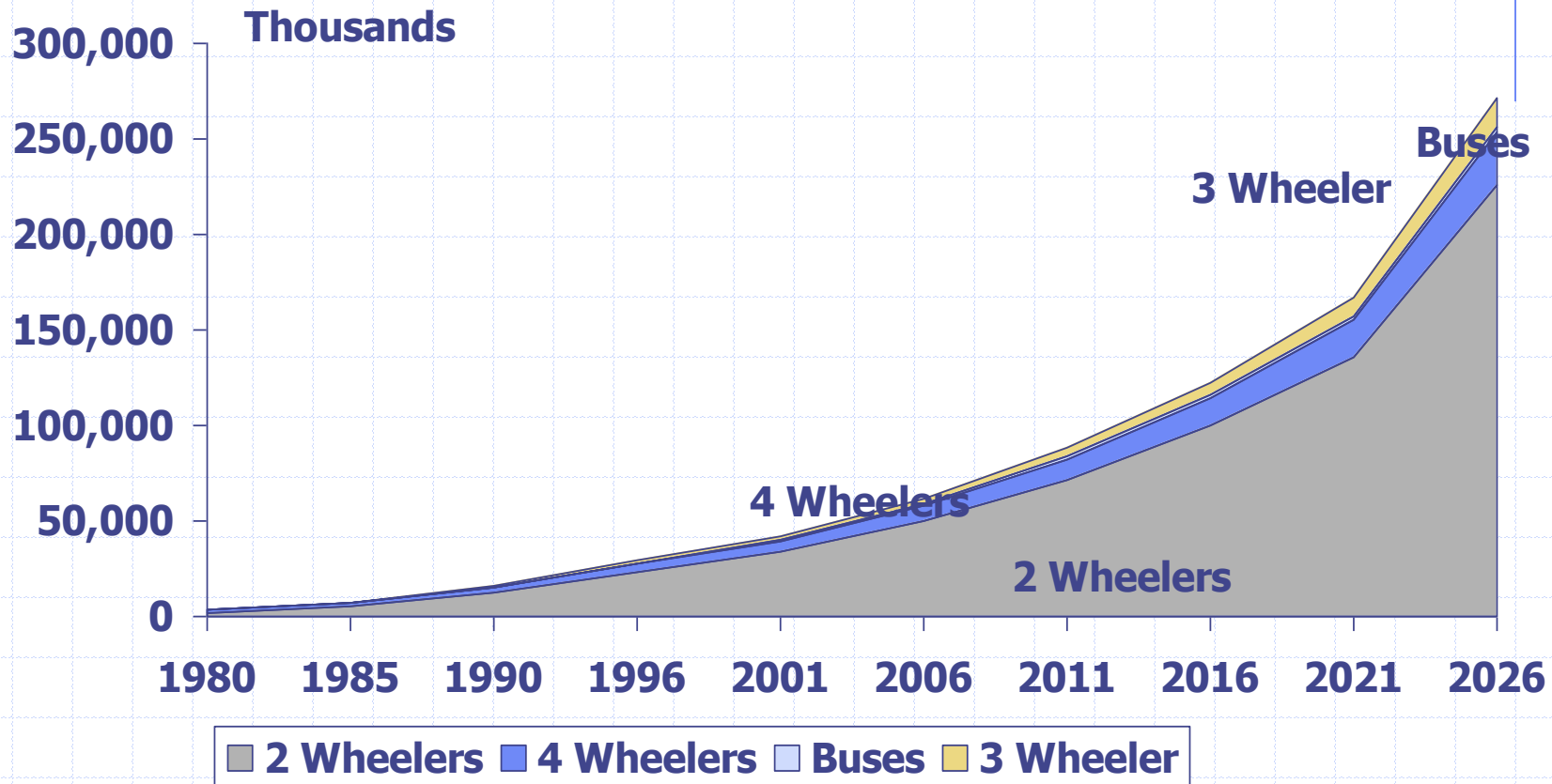
Vehicle Ownership

Cities	Vehicles Per 1000 Persons	Per capita income (US \$)
London (1990)	356	\$22,000
Paris (1990)	383	\$34,000
Tokyo (1990)	266	\$37,000
Delhi (1998)	200	\$850
Shanghai (2000)	69	\$4,000

Urban travel demand projections



Vehicular Growth



Policy response in India

- ◆ Largely technology based response
 - *Vehicle emissions*
 - *Fuel quality*
 - *Mandating alternative fuels*
- ◆ Inadequate attention to demand management
 - *NMT discriminated against in resource*
 - *Land use development haphazard, not in sync with transport planning*
 - *Public transport not encouraged, personal vehicles have become more accessible*

Emission norms

New vehicles

- ◆ 1991: First mass emission norms (CO & HC)
- ◆ 1995: Catalytic converters for cars in 4 metros
- ◆ 1996: Further tightening of norms (CO, HC & NO_x)
- ◆ 2000: Bharat Stage I (~EUROI)
- ◆ 2001: Bharat Stage II in Delhi (~EURO II)
- ◆ Sustained progress!!!

In use vehicles

- ◆ 1989: Idle emission regulation
- ◆ 2004: Some tightening
- ◆ Piece meal efforts

No Statutory Maintenance & Fuel Efficiency Requirements !!!

Government's Auto Fuel Policy

Category	Bharat Stage II April 2005	Bharat Stage III April 2010	Bharat Stage IV
2 & 3 wheelers	Entire country	Entire country (preferably from April 2008)	
All other new vehicles	Entire country	Entire country	
	April 2003 7 cities other than metros	April 2005 11 major cities	April 2010 11 major cities

Need for an effective system for in use vehicles

- ◆ Norms very lax
 - *No consistent improvement*
- ◆ Limited coverage
 - *Only idle tests for CO (petrol)*
 - *Free acceleration tests for smoke (diesel)*
- ◆ Limited regulatory ability
 - *Equipment not calibrated*
 - *No auditing/ training*
- ◆ **Weak maintenance system**

Synergizing Objectives

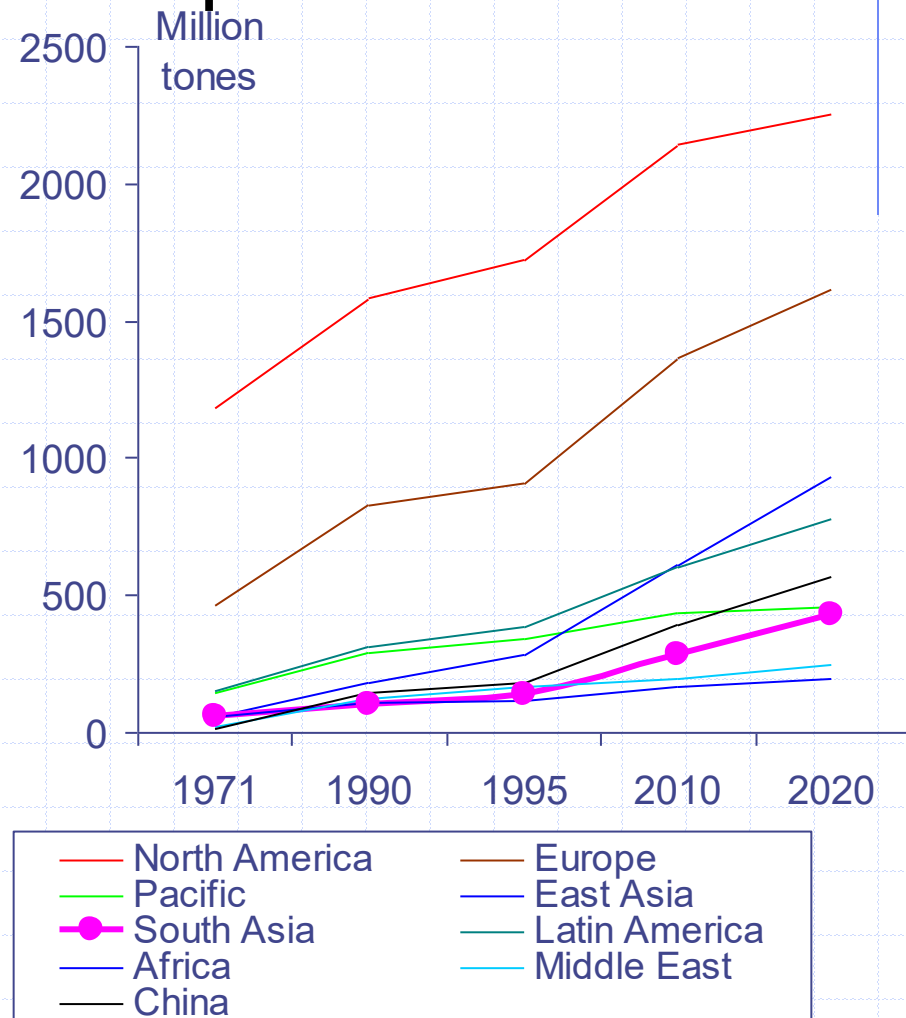
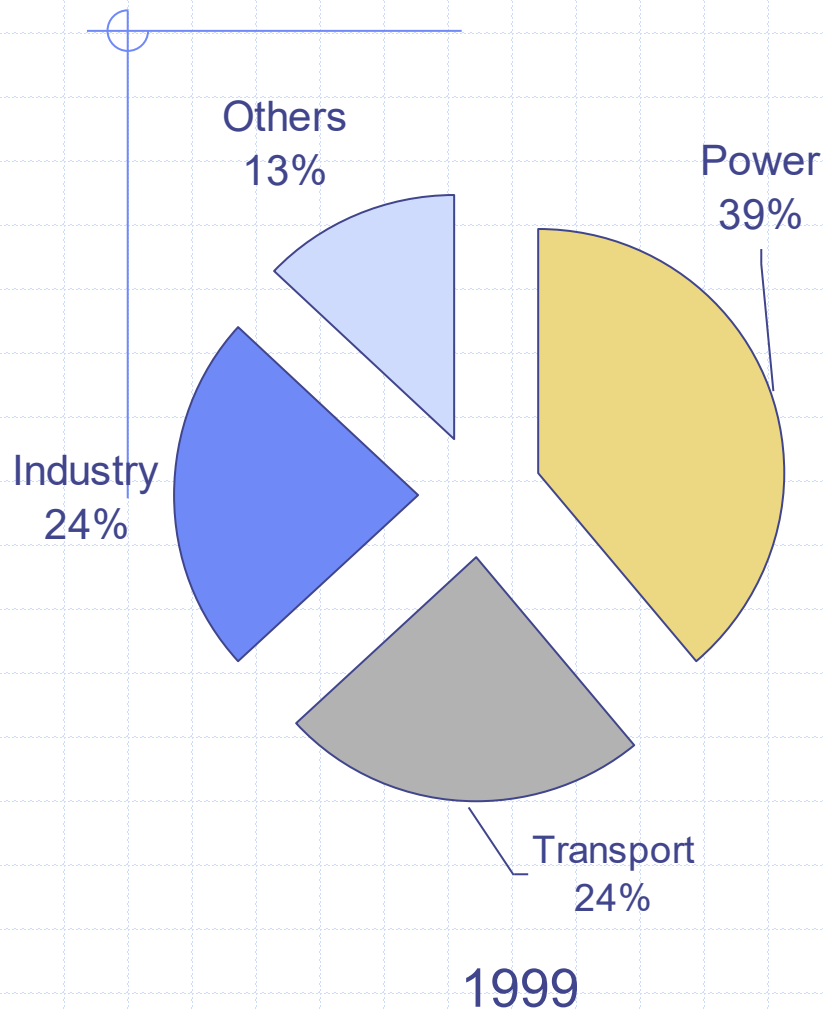
◆ Global

- *Significant potential for GHG mitigation in transport in India*

◆ National

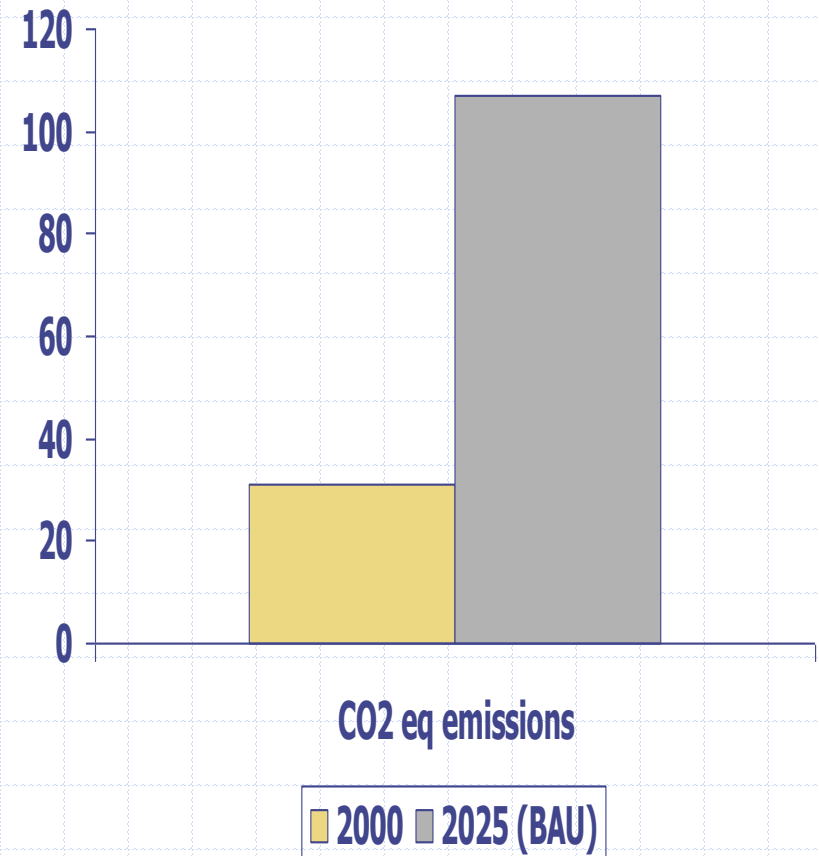
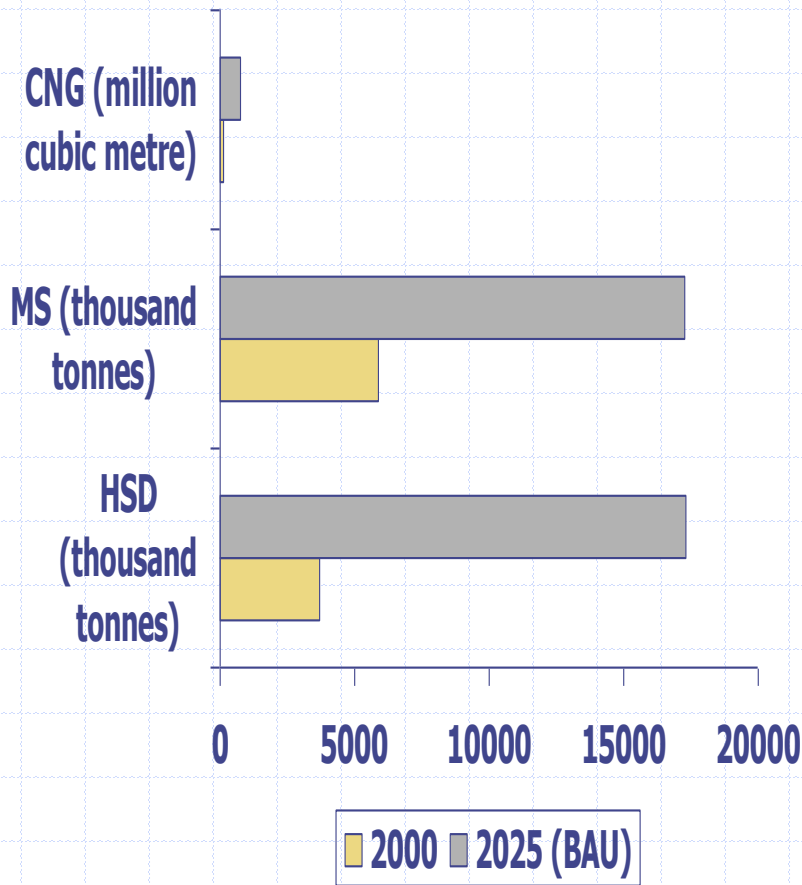
- *Positive impacts on local air quality, health & safety*
- *High priority for urban centres*
- *Revenue from carbon credits*

GHG emissions from transport

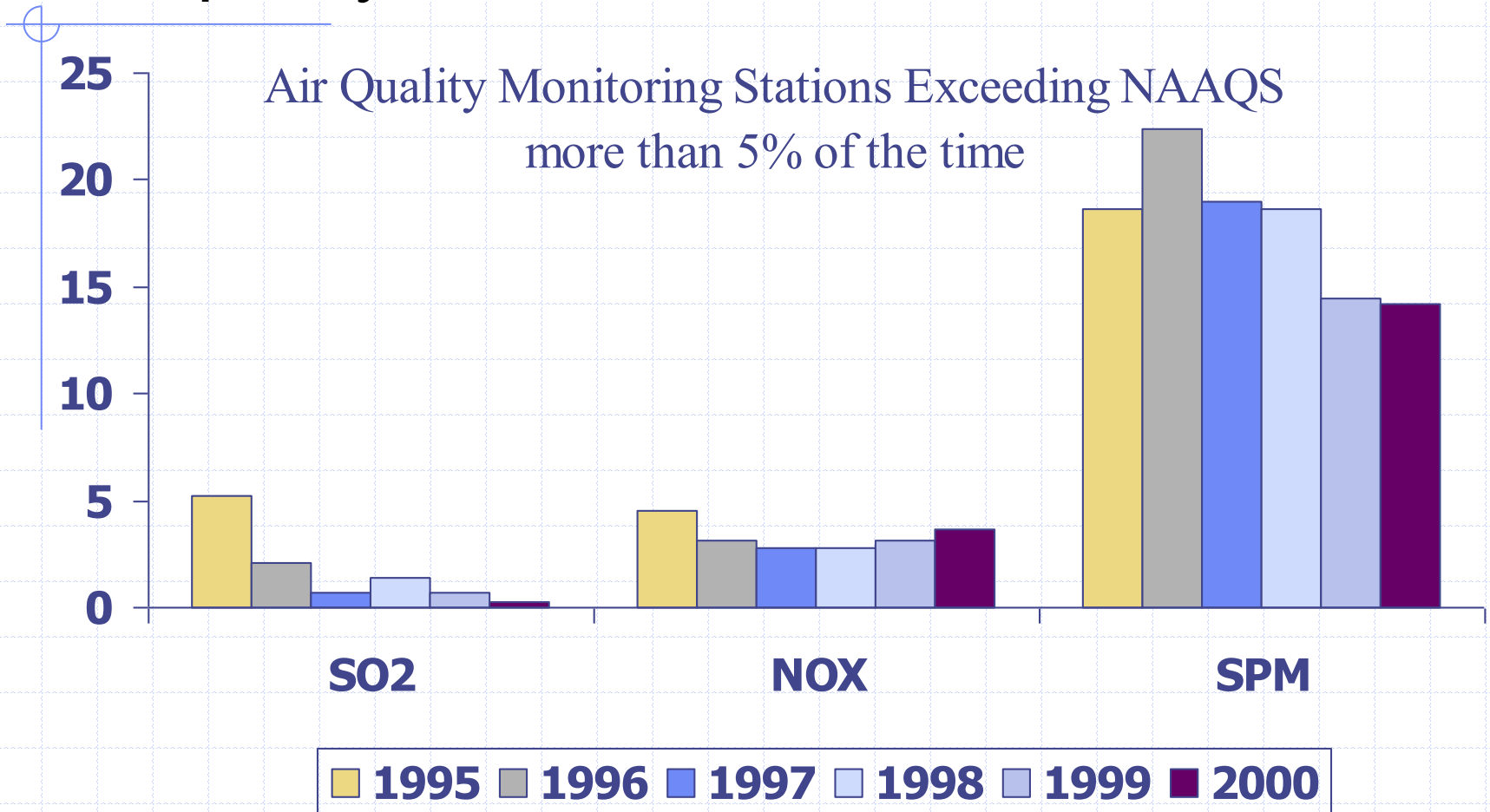


OECD & IEA 2001

Fuel consumption & GHG emissions in Urban India

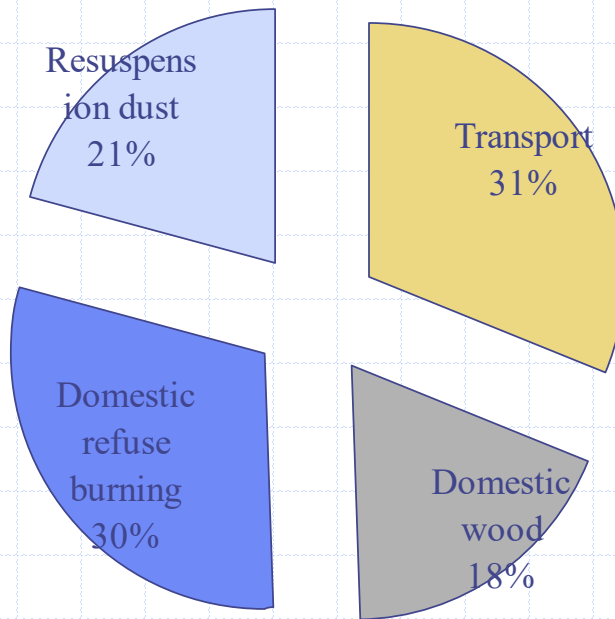


Air quality is a concern

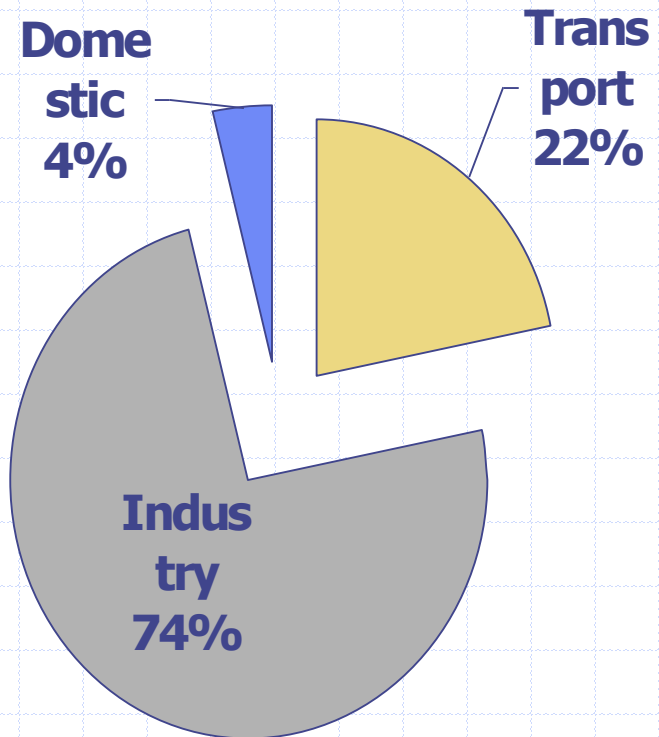


Transport contributes 3-22% of PM emissions

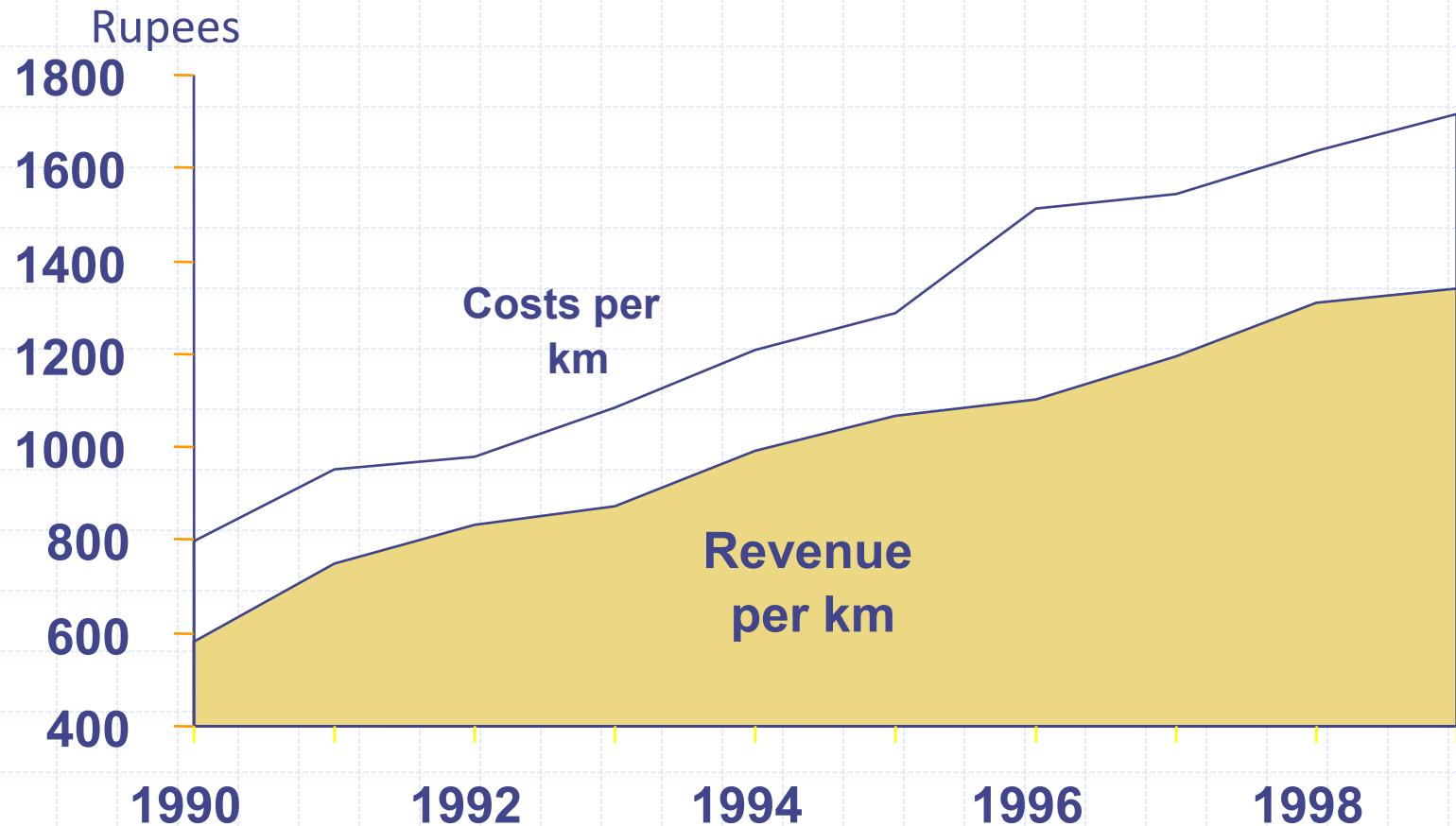
UrbAir Mumbai (WB 1995)



Delhi (CPCB 1994)



Financial performance of public transport operators in India



Barriers

◆ Baselines

- *data availability*
- *uncertain sectoral forecasts*

◆ Leakages

- *Fuel efficiency Vs VOCs*

◆ Monitoring and verification costs

- *dispersed and large number of mobile sources*

◆ Additionality

- *Mandated use of alternative fuels*
- *Comprehensive I&C regime proposed*

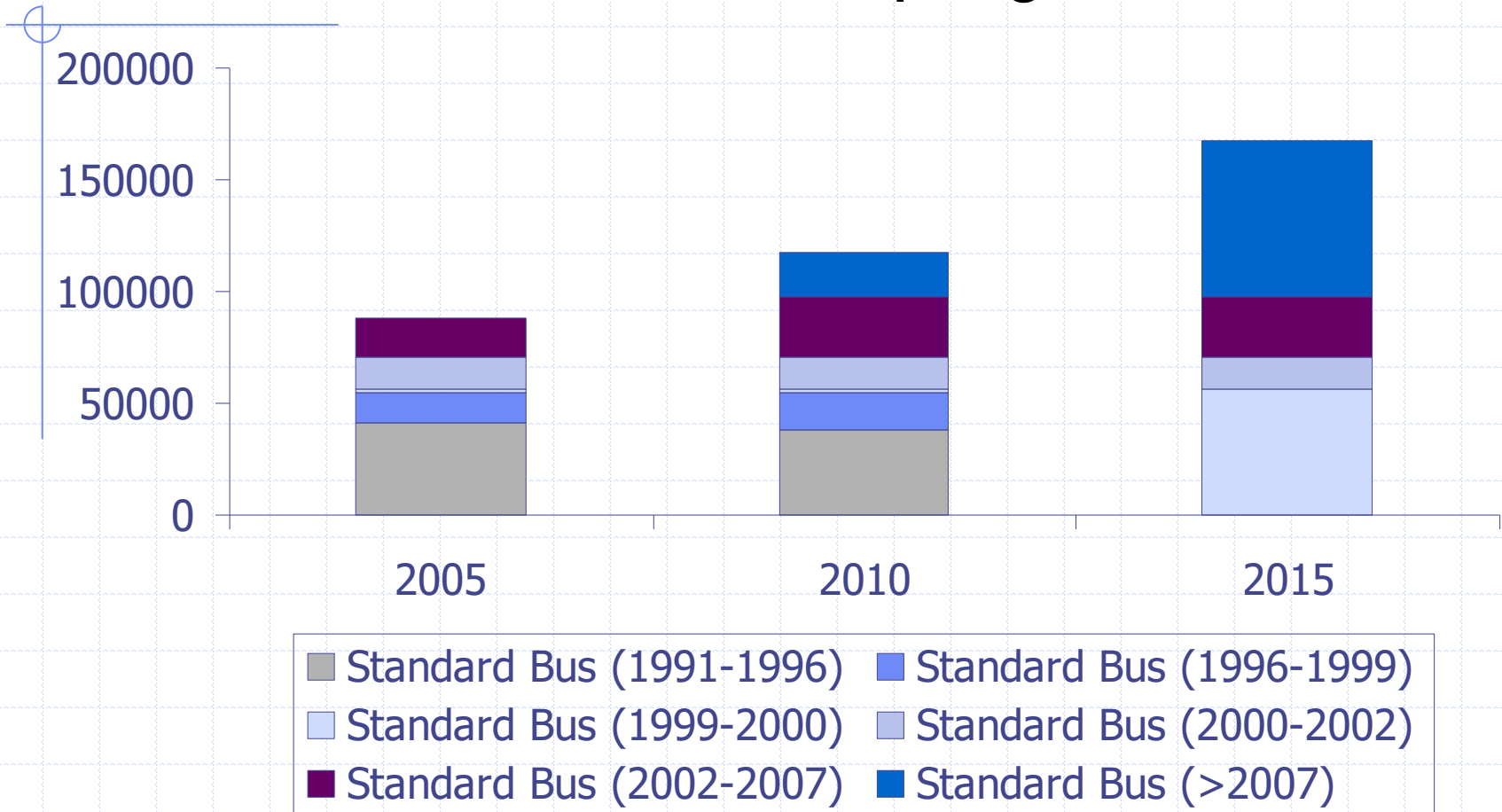
Project description

- ◆ Existing maintenance regime inadequate and primitive
 - *Even minor improvements would have large benefits*
- ◆ GHG savings from
 - *Better maintenance*
 - *Result: Improved energy efficiency. Hence*
 - ◆ GHG savings, Local emissions benefits
- ◆ Impact: Range estimate of GHG reduction
 - *5% improvement in fuel efficiency in US (Das et al 2001)*
 - *5.5% improvement in Jakarta (Cornie)*
 - *12.6% improvement in fuel efficiency (USEPA IM240 Program)*
 - *10-20% improvement in fuel efficiency in India (Das et al)*

Project structure

- ◆ Project proponent: Thane Municipal Transport Undertaking
 - *Statutory public bus company*
 - *Owned by Thane Municipal Corporation*
- ◆ Mandate: Provide transport services to Thane
- ◆ Activities
 - *Bus operation*
 - *Depots and terminals*
 - *Bus maintenance*

Bus fleet covered in the program



Maintenance checklist

◆ Visual safety tests

- *Steering*
- *Chassis*
- *Fuel tank & piping*
- *Engine mountings*
- *Battery terminals etc*
- *Seatbelts*
- *Lighting*
- *Oil leakages*
- *Leaf springs...*

◆ Automated safety tests

- *Headlamp beam*
- *Brakes*
- *Sideslip*

◆ Emissions

- *Volumetric concentration for gasoline*
- *Opacity for diesel*

Design issues

◆ Project boundary

- *Government owned bus company. Reduces*
 - ◆ institutional complexity
 - ◆ data requirements, thus project development costs

◆ Operational lifetime

- *Life of equipment = 15 years*

◆ Crediting period: 10 years

- *Reduced uncertainty*

◆ Monitoring & verification

- *Fuel consumption already monitored daily*
- *Hence, no expense on monitoring*

Emission savings: Methodology

- ◆ GHG emissions

$$(Vehicle_kms) \times (Emissions/km)$$

- ◆ Vehicle kms

$$(Vehicles) \times (Utilisation)$$

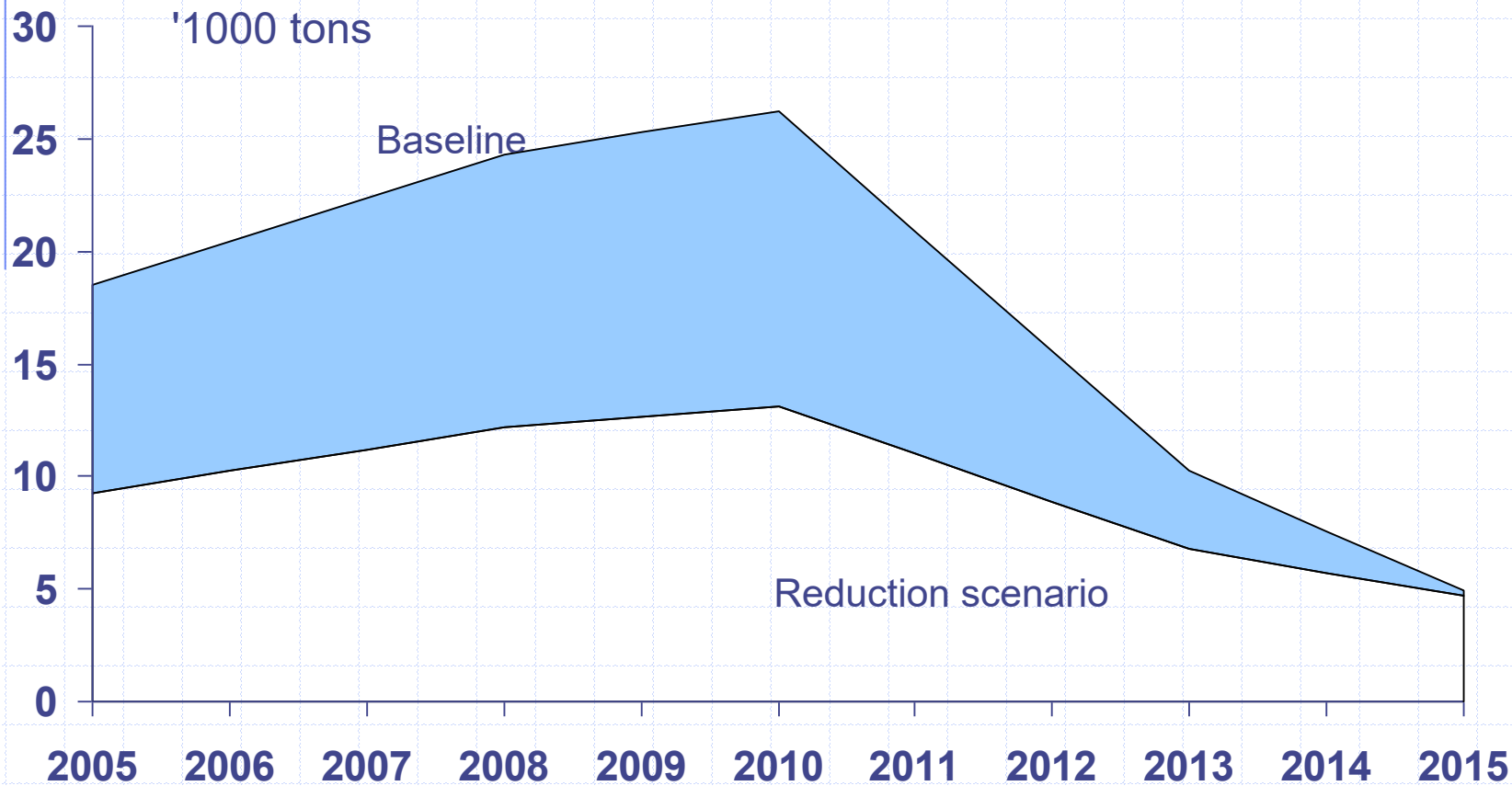
- ◆ Baseline projections based on

- *Existing travel trends*
- *Expected changes in emission factors, fuel efficiencies*
 - ◆ Increasingly stringent norms
 - ◆ Improved technology

- ◆ Sustained improvements in efficiency: 5%

- ◆ Reference years: 2005 & 2015

GHG savings



Project costs

- ◆ Net present value of project
 - — *US\$ 0.13 million*
- ◆ Project cost
 - *US\$ 0.54 million*
- ◆ Cost of CO₂
 - ~ *US\$ 8.5*
- ◆ Substantial local benefits

Transport CDM projects: Lessons

◆ Baselines

- *Carry out pilot tests to estimate fuel efficiency gains*

◆ Leakages

- *Captive fleet with fixed mobility*
- *Independent of VOCs, exogenously determined demand*

◆ Monitoring and verification costs

- *Data already monitored independently*
- *Preexisting statutory auditing: CAG*

◆ Additionality

- *Comprehensive I&C regime proposed in India*
- *But no maintenance guidelines, requirements*

Structure of a CDM project

The Executive Board

◆ Baselines

- *Additionality*
- *Leakages*

◆ Monitoring & verification

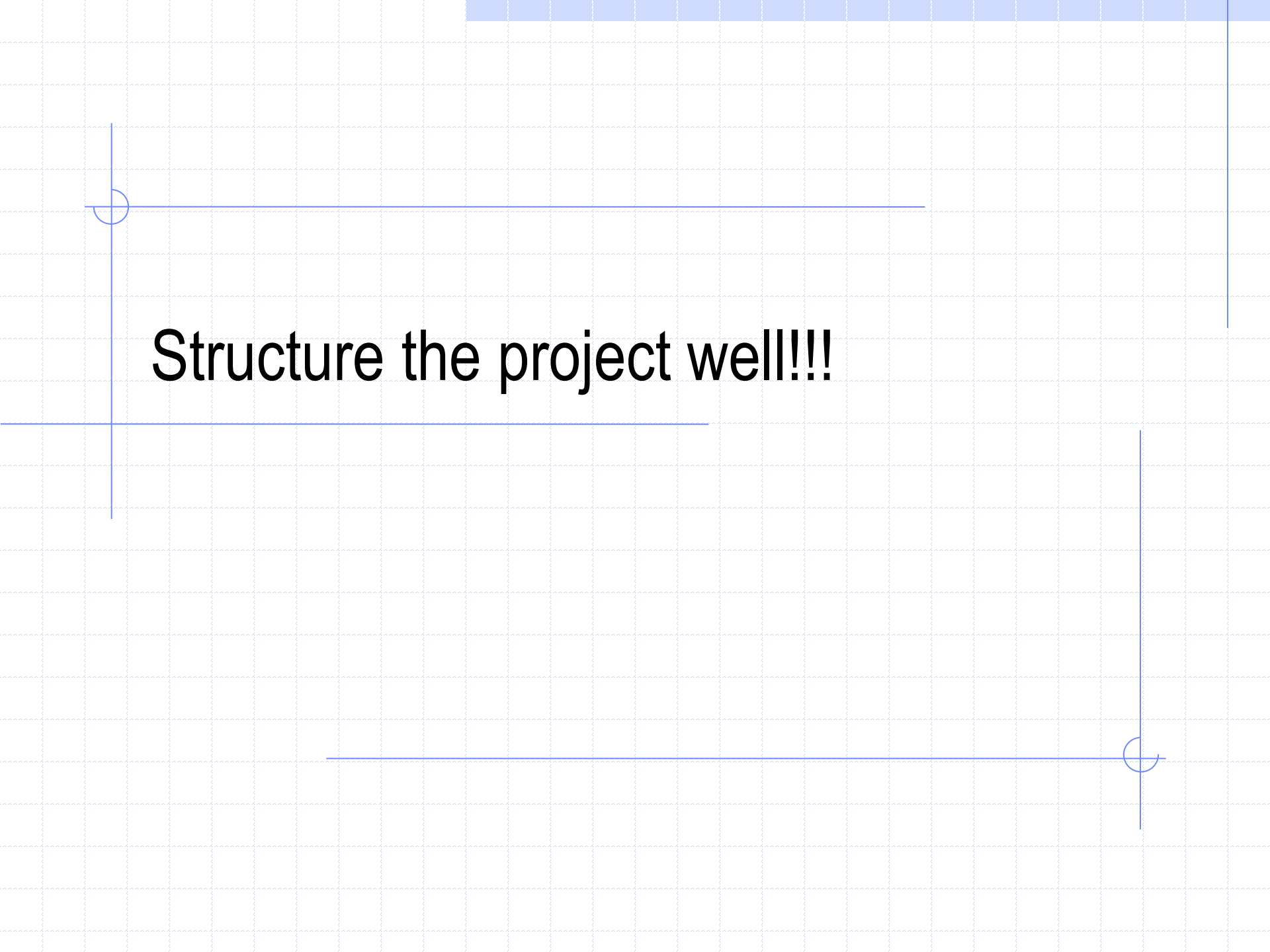
CER market

◆ Price of CERs

- *Total CERs*
- *Cost of project*

◆ Cost issues

- *Equity/Debt Vs revenue streams*
- *CER revenues only as revenue streams!!!*

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Structure the project well!!!